PG No. 27 D_{6h} $6/mmm$ [hexagonal]

* character table

D_{6h}	$1(1)$	$2_{001}(1)$	$2_{100}(3)$	$2_{120}(3)$	$3^+_{001}(2)$	$6^+_{001}(2)$	$-1(1)$	$m_{100}(3)$	$m_{001}(1)$	$m_{120}(3)$	$-3^+_{001}(2)$	$-6^+_{001}(2)$
A_{1g}	1	1	1	1	1	1	1	1	1	1	1	1
A_{2g}	1	1	−1	−1	1	1	1	−1	1	−1	1	1
B_{1g}	1	−1	−1	1	1	−1	1	−1	−1	1	1	−1
B_{2g}	1	−1	1	−1	1	−1	1	1	−1	−1	1	−1
E_{1g}	2	−2	0	0	−1	1	2	0	−2	0	−1	1
E_{2g}	2	2	0	0	−1	−1	2	0	2	0	−1	−1
A_{1u}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
A_{2u}	1	1	−1	−1	1	1	−1	1	−1	1	−1	−1
B_{1u}	1	−1	−1	1	1	−1	−1	1	1	−1	−1	1
B_{2u}	1	−1	1	−1	1	−1	−1	−1	1	1	−1	1
E_{1u}	2	−2	0	0	−1	1	−2	0	2	0	1	−1
E_{2u}	2	2	0	0	−1	−1	−2	0	−2	0	1	1

* polar ↔ axial conversion

A_{1g} (A_{1u}) A_{2g} (A_{2u}) B_{1g} (B_{1u}) B_{2g} (B_{2u}) E_{1g} (E_{1u}) E_{2g} (E_{2u}) A_{1u} (A_{1g}) A_{2u} (A_{2g}) B_{1u} (B_{1g}) B_{2u} (B_{2g}) E_{1u} (E_{1g}) E_{2u} (E_{2g})

* symmetric product

	A_{1g}	A_{2g}	B_{1g}	B_{2g}	E_{1g}	E_{2g}	A_{1u}	A_{2u}	B_{1u}	B_{2u}	E_{1u}	E_{2u}
A_{1g}	A_{1g}	A_{2g}	B_{1g}	B_{2g}	E_{1g}	E_{2g}	A_{1u}	A_{2u}	B_{1u}	B_{2u}	E_{1u}	E_{2u}
A_{2g}		A_{1g}	B_{2g}	B_{1g}	E_{1g}	E_{2g}	A_{2u}	A_{1u}	B_{2u}	B_{1u}	E_{1u}	E_{2u}
B_{1g}			A_{1g}	A_{2g}	E_{2g}	E_{1g}	B_{1u}	B_{2u}	A_{1u}	A_{2u}	E_{2u}	E_{1u}
B_{2g}				A_{1g}	E_{2g}	E_{1g}	B_{2u}	B_{1u}	A_{2u}	A_{1u}	E_{2u}	E_{1u}
E_{1g}					$A_{1g}+E_{2g}$	$B_{1g}+B_{2g}+E_{1g}$	E_{1u}	E_{1u}	E_{2u}	E_{2u}	$A_{1u}+A_{2u}+E_{2u}$	$B_{1u}+B_{2u}+E_{1u}$
E_{2g}					$A_{1g}+E_{2g}$		E_{2u}	E_{2u}	E_{1u}	E_{1u}	$B_{1u}+B_{2u}+E_{1u}$	$A_{1u}+A_{2u}+E_{2u}$
A_{1u}							A_{1g}	A_{2g}	B_{1g}	B_{2g}	E_{1g}	E_{2g}
A_{2u}								A_{1g}	B_{2g}	B_{1g}	E_{1g}	E_{2g}
B_{1u}									A_{1g}	A_{2g}	E_{2g}	E_{1g}
B_{2u}										A_{1g}	E_{2g}	E_{1g}
E_{1u}											$A_{1g}+E_{2g}$	$B_{1g}+B_{2g}+E_{1g}$
E_{2u}												$A_{1g}+E_{2g}$

* anti-symmetric product

A_{1g}	A_{2g}	B_{1g}	B_{2g}	E_{1g}	E_{2g}	A_{1u}	A_{2u}	B_{1u}	B_{2u}	E_{1u}	E_{2u}
–	–	–	–	A_{2g}	A_{2g}	–	–	–	–	A_{2g}	A_{2g}